Unrestricted State–Stack Pareto Frontiers and Bounded-Push Tradeoffs for Rational Binary Balance Languages

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Abstract

For coprime integers $p, q \geq 1$, let

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}.$$

A representation-independent baseline theorem gives the exact state minimum $\max(p, q)$ for deterministic pushdown recognition in a broad regular-target model and a matching right-endmarker construction with three total stack symbols. The missing two-resource question is whether shrinking the stack alphabet forces additional finite control when the competing recognizer is otherwise arbitrary.

We prove that it does. In a permanent-bottom model with exactly one work symbol, every deterministic ε-pushdown recognizer of $L_{p,q}$ has at least $p+q$ states, with no residual-faithful encoding, homogeneous-tail, realtime, or bounded-replacement hypothesis. Together with known constructions, this yields the exact unrestricted right-endmarker Pareto frontier

$$\mathcal{F}_{p,q} = \{(p + q, 2), (\max(p, q), 3)\}.$$

The new lower bound is obtained by aligning the positive and negative protected-pump cut families of an arbitrary recognizer and using a finite-depth unary descent lemma to force all $p + q$ residue phases onto one physical unary stack.

We then charge replacement length as a third resource inside the explicitly typed raw residual-conjugacy category. If every transition replaces the visible top by a word of length at most $B \geq 2$, the exact one-work-symbol state minimum is

$$\max(p, q) + \max\left(\min(p, q), \left\lceil \frac{\max(p, q)}{B - 1} \right\rceil\right).$$

The proof uses a fixed residual-+1 word to induce weighted finite-state cycles on the two deep unary tails; their weights have opposite signs, while bounded push and one-symbol pop locality give sharp cycle-capacity inequalities. This yields an exact raw unary state–push staircase, the critical push threshold for recovering the unbounded unary optimum, and an inverse state-budget–to–push law. Separately, the optimal three-symbol M-state point survives replacement-at-most-two by a uniquely forced ε-suffix compiler. That compiler is a normalized macro-step realization rather than a raw one-transition residual conjugacy, so the two bounded statements are kept in their correct categories.

1 Introduction

The present paper continues the resource-complexity line generated by the completion-aware 2:1 deterministic pushdown construction and its coprime p:q generalization. The signed residual

$$\rho(w) = qN_1(w) - pN_0(w)$$

classifies right quotients of $L_{p,q}$ and therefore supplies a language-intrinsic semantic coordinate, while physical pushdown realizations of that coordinate can vary substantially.

From the original 2:1 machine to the resource problem. Figure 1 records the historical machine from which this resource line emerged. In the original completion-aware 2:1 DPDA, finite control and the literal stack jointly carried the remaining completion demand, while end-of-input rules materialized the bounded control offset into a completion word. Passing to coprime $p{:}q$ isolated the residual ρ from that particular physical encoding. The later unrestricted state theorem then showed that the finite-control cost $\max(p, q)$ survives even when the opponent is allowed to abandon the residual-faithful representation completely. The present paper asks the next resource question: what changes when the stack alphabet, and then the replacement length, are charged as resources as well?

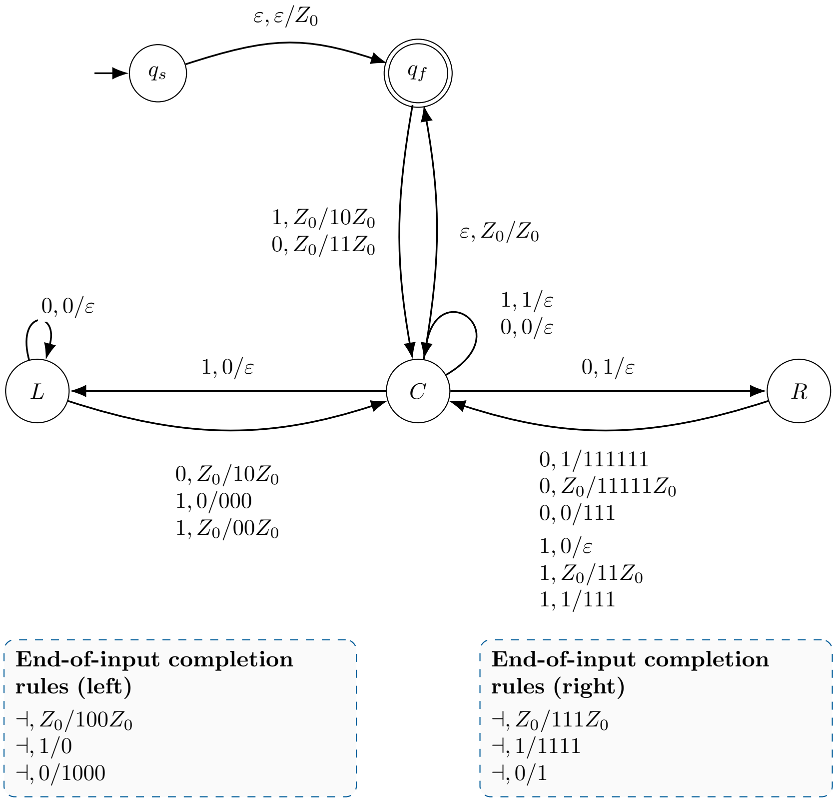

Figure 1: The original completion-aware 2:1 DPDA, in the empty-word-included formal variant. The figure records the provenance of the resource program: the later $p{:}q$ residual and the unrestricted/Pareto theorems do not assume this transition table or its physical encoding.

The conceptual progression relevant here is therefore

$$\boxed{\begin{array}{c}\text{original 2:1} \longrightarrow p{:}q \text{ residual/completion} \longrightarrow Q_{\min}(L_{p,q}) = \max(p, q) \\ \downarrow \\ \text{state–stack Pareto frontier} \longrightarrow \text{bounded-push deformation}\end{array}}$$

The first two arrows explain the origin; the final two are the resource questions developed in this paper.

The self-contained baseline theorem proved in Appendix B removes representation assumptions entirely and shows that every deterministic ε-pushdown recognizer in the regular-target model satisfies

$$|Q| \geq M := \max(p, q),$$

with a matching right-endmarker construction using three total stack symbols. That theorem settles the state coordinate after the stack alphabet is left free. It does not determine the full state/stack antichain: in particular, it does not say how many states are necessary if the total stack alphabet is forced down to two symbols, one of which is the permanent bottom marker.

The first contribution of this paper is to close exactly that gap. The lower-bound opponent may use arbitrary deterministic ε-moves, noncanonical intermediate configurations, and arbitrary finite replacement words. The only additional resource restriction is that the non-bottom work alphabet is unary. We show that this forces $p+q$ states. Numerically the resulting antichain coincides with an earlier raw residual-faithful frontier, but the theorem class is stronger: here the opponent is an arbitrary recognizer in the stated permanent-bottom model.

The second contribution studies bounded replacement. This question must be typed more carefully. In the unrestricted recognizer class, a bounded-push machine may distribute semantic information across history-dependent representatives and forced normalization. We therefore prove the exact state–push spectrum only on the one-work-symbol raw residual-conjugacy branch. We also prove, in the broader normalized macro-step model, that the M-state three-symbol point survives replacement-at-most-two with uniquely forced ε-suffix chains. These two bounded results are intentionally not merged into one "raw" frontier.

2 Model and baseline facts

Fix coprime $p, q \geq 1$. A deterministic pushdown automaton has finite control Q, input alphabet $\{0, 1\}$ together with a mandatory right endmarker $\dashv$ for the exact upper-bound statements, and stack alphabet Γ containing a permanent bottom marker Z_0. The marker occurs exactly once, at the bottom, and is neither popped nor copied. The stack top is written on the left in this paper. A transition rewrites only the visible top symbol. Deterministic ε-moves are allowed unless a theorem explicitly says raw or realtime.

The bottom marker is counted in $|\Gamma|$. Thus $|\Gamma| = 2$ means

$$\Gamma = \{A, Z_0\}$$

for one work symbol A.

Definition 2.1 (Unrestricted resource frontier)**.** Let $\mathcal{F}_{p,q}$ be the Pareto-minimal set of pairs $(|Q|, |\Gamma|)$ among deterministic right-endmarker pushdown recognizers of $L_{p,q}$ in the permanent-bottom model.

We use two baseline facts. First, right quotients are exactly classified by the residual: two prefixes have the same accepting continuation set if and only if they have the same value of ρ. Consequently every reachable physical configuration has a well-defined residual label. Second, the representation-independent state theorem gives

$$|Q| \geq M = \max(p, q), \tag{1}$$

for arbitrary deterministic ε-pushdown recognizers in the broader regular-target model, and the bound is attained under the right-endmarker convention with three total stack symbols.

The lower-bound proof of (1) supplies the following interface, which we restate because the new unary argument begins there.

Lemma 2.2 (Protected unary pump interface)**.** *On the live computation over 1^ω, there exist a state r, a visible stack symbol X, an integer $c \geq 1$, a nonempty word U, and a lower context η such that*

$$(r, X\eta) \stackrel{1^c}{\Longrightarrow} (r, XU\eta),$$

where X is the visible top symbol and η is the untouched lower context. The macro includes all forced ε-moves and never drops below the height of $X\eta$. It is context-independent while η remains hidden. It therefore iterates as $(r, XU^k\eta)$ and creates protected cuts $H_k = U^k\eta$.

Lemma 2.3 (Phase exposure interface). *From the positive pump of Theorem 2.2, one obtains p synchronized phase families. For each $j \in \{0, \dots, p-1\}$ there are constants z_j and states s_j such that, for every pump index k, the first exposure of the common cut H_k along the 0^ω continuation occurs at*

$$(s_j, H_k),$$

after z_j consumed zeros. The states $s_0, \dots, s_{p-1}$ are pairwise distinct. The symmetric negative construction gives q pairwise distinct phases.

Permanent refusal to expose a protected cut is excluded by the same two-pump diagonal argument used in the state lower bound: a second context-independent pump would produce a two-parameter reachable family whose acceptance under one fixed continuation is a nonregular linear diagonal, contradicting regularity of fixed-continuation predecessor stack languages.

3 Finite-depth descent in a unary stack

Assume now $\Gamma = \{A, Z_0\}$. The protected pump word in Theorem 2.2 is necessarily $U = A^u$ for some $u \geq 1$, so after rebasing the positive cuts have the form

$$H_k = A^{a+uk} Z_0.$$

Lemma 3.1 (Finite-depth unary descent). *Fix an integer $d \geq 0$. For each positive phase j, after discarding finitely many initial pump indices, there exist an integer $e_{j,d} \geq 0$ and a state $s_{j,d}$, independent of k, such that, writing*

$$H_k = A^d L_k,$$

one has

$$(s_j, A^d L_k) \xRightarrow{0^{e_{j,d}}} (s_{j,d}, L_k)$$

for every sufficiently large k.

Proof. Before L_k is exposed, the relative configuration above it is the fixed pair (s_j, A^d). Therefore the entire deterministic raw history above the cut is independent of k. Either the relative stack empties after one fixed history for all sufficiently large k, or it never empties for any such k.

Assume the second alternative. Every finite zero-prefix remains live, raw recurrence is impossible because a repeated physical configuration would carry two residuals, and the relative height therefore escapes. Applying the protected-pump construction to this relative 0^ω computation yields a second context-independent pump. Because the stack alphabet has only the work symbol A above Z_0, write its height increment as $v \geq 1$ and let it consume $e \geq 1$ zeros. After the fixed relative prefix that reaches the second pump mode, iteration of the original 1-pump and of this new 0-pump therefore produces, for all sufficiently large k, ℓ, configurations with one fixed control state, residual

$$B + kcq - \ell ep,$$

and unary stack

$$A^{H+uk+v\ell} Z_0$$

for fixed integers B, H, where the original pump has height increment $u \geq 1$ and consumes $c \geq 1$ ones.

Now compare the reachable parameter pairs $(k + v, \ell)$ and $(k, \ell + u)$. Their stack heights are identical:

$$H + u(k + v) + v\ell = H + uk + v(\ell + u),$$

so, in a unary stack, the complete physical stacks are identical; the control state is identical as well. Their residuals, however, differ by

$$vcq + uep > 0.$$

This contradicts residual rigidity: one reachable physical configuration cannot have two distinct accepting-continuation quotients. Hence the non-exposure alternative is impossible, and the fixed depth is eventually removed. $\square$

Corollary 3.2 (Phase separation survives descent)**.** *For fixed d, all descended phase states are pairwise distinct. In particular,*

$$s_{j,d} \neq s_{t,d} \qquad (0 \leq j < t \leq p-1).$$

Proof. All descended configurations have the same stack L_k. Equality of the states for phases $j \neq t$ would therefore identify two reachable physical configurations. Their residual difference has the form

$$q(j-t) - p(e_{j,d} - e_{t,d}),$$

which cannot vanish because $\gcd(p, q) = 1$ and $0 < |j-t| < p$. $\square$

The symmetric statement holds for the q negative phases. For fixed descent depth, the positive residuals tend to $+\infty$ with the pump index and the negative residuals tend to $-\infty$.

4 Exact unrestricted one-work-symbol complexity

Let the positive and negative protected cut heights be

$$h_k^+ = a + uk, \qquad h_\ell^- = b + v\ell,$$

with $u, v \geq 1$. Put $g = \gcd(u, v)$. Choose large k_0, ℓ_0 and set

$$k_n = k_0 + \frac{v}{g}n, \qquad \ell_n = \ell_0 + \frac{u}{g}n.$$

Then the height difference

$$h_{k_n}^+ - h_{\ell_n}^- = D$$

is constant.

Theorem 4.1 (Unrestricted one-work-symbol lower bound)**.** *Every deterministic ε-pushdown recognizer of $L_{p,q}$ in the permanent-bottom model with exactly one work symbol satisfies*

$$\boxed{|Q| \geq p + q.}$$

No raw residual conjugacy, homogeneous-tail, realtime, or bounded-replacement hypothesis is assumed.

Proof. Suppose $D \geq 0$. Apply Theorem 3.1 with depth D to all positive phases and leave the negative family at depth zero. At index n, all $p+q$ phase configurations now have the identical physical stack

$$A^{h_{\ell_n}^-} Z_0.$$

By Theorem 3.2, the p positive states are pairwise distinct; the symmetric argument makes the q negative states pairwise distinct. For sufficiently large n, every descended positive residual is positive and every negative residual is negative. Hence no positive state can equal a negative state, because equality together with the common stack would identify two reachable physical configurations carrying different residuals. Thus at least $p+q$ states are required. The case $D < 0$ is symmetric. $\square$

The raw construction of Theorem 5.2 at any $B \geq 1 + \lceil M/m \rceil$ uses exactly $p+q$ states and is ε-free, so the bound is sharp.

Theorem 4.2 (Exact unrestricted state–stack frontier)**.** *For every coprime $p, q \geq 1$,*

$$\boxed{\mathcal{F}_{p,q} = \{(p+q, 2), (M, 3)\}, \qquad M = \max(p, q).}$$

Proof. With no work symbol, only finitely many physical configurations are available, so the recognized language is regular. Thus $|\Gamma| = 1$ is impossible. At $|\Gamma| = 2$, Theorem 4.1 gives the sharp lower bound $p+q$. For every alphabet size at least three, the baseline theorem (1) gives $|Q| \geq M$, while a three-symbol construction attains $(M, 3)$. Every point with at least three stack symbols is dominated by $(M, 3)$, and the two displayed points are incomparable because $M < p + q$. □

For the motivating 2:1 language,

$$\mathcal{F}_{2,1} = \{(3, 2), (2, 3)\}.$$

5 Bounded replacement in the raw residual-conjugacy category

We now type the representation more strongly.

Definition 5.1 (Raw B-bounded unary residual conjugacy)**.** Let $B \geq 2$. A raw B-bounded unary residual conjugacy is an injective map

$$\eta : \mathbb{Z} \to Q \times A^* Z_0$$

such that

$$\delta(\eta(r), 0) = \eta(r - p), \qquad \delta(\eta(r), 1) = \eta(r + q)$$

by one ordinary binary transition, with no normalization step on encoded residual configurations, and every replacement word has length at most B.

This category is narrower than the unrestricted opponent of Theorem 4.1; the gain is that bounded-push effects become exactly measurable.

5.1 A matching bounded construction

Assume $p \geq q$ and define

$$b_B = \max\left(q, \left\lceil \frac{p}{B-1} \right\rceil\right).$$

For $r \geq 0$, write $r = pk + s$, $0 \leq s < p$, and encode

$$\eta(r) = (P_s, A^k Z_0).$$

For $r < 0$, write uniquely

$$r = u - b_B h, \qquad 0 \leq u < b_B, \qquad h \geq 1,$$

and encode

$$\eta(r) = (N_u, A^{h-1} Z_0).$$

The positive transitions use the usual quotient/remainder updates modulo p. On the negative side, input 1 changes the quotient height by at most one because $b_B \geq q$, while input 0 increases it by at most $B - 1$ because

$$b_B \geq \left\lceil \frac{p}{B-1} \right\rceil.$$

The boundary identities are the corresponding Euclidean decompositions. Hence every replacement word has length at most B.

Proposition 5.2 (Bounded raw upper construction)**.** *For $p \geq q$ and $B \geq 2$, there is an ε-free raw B-bounded unary residual conjugacy with*

$$|Q| = p + b_B.$$

By symmetry, for arbitrary coprime p, q,

$$|Q| \leq M + \max\left(m, \left\lceil \frac{M}{B-1} \right\rceil\right), \quad m = \min(p, q).$$

5.2 Weighted unit cycles on the two deep tails

Choose nonnegative integers a, b satisfying

$$qa - pb = 1$$

and fix the word

$$w = 1^a 0^b.$$

For all sufficiently large positive residuals, every prefix residual while reading w remains positive. For sufficiently negative residuals, the same is true with "negative" in place of "positive". Raw tail escape makes all corresponding encoded stacks nonempty. Thus on both deep tails the same word w induces one deterministic weighted map on top-A modes,

$$G : (s, h) \mapsto (g(s), h + \omega(s)),$$

and G sends $\eta(r)$ to $\eta(r+1)$.

Lemma 5.3 (Signed tail cycles)**.** *The positive deep tail is eventually periodic on a directed state cycle C_+ of length L_+ and total height weight $+c_+$ for some $c_+ \geq 1$. The negative left-infinite tail is eventually periodic on a directed cycle C_- of length L_- and total height weight $-c_-$ for some $c_- \geq 1$. The two cycles are state-disjoint.*

Proof. Finite-state determinism gives eventual cycles. A zero total weight would repeat the same state and stack height at residuals differing by the cycle length, contradicting injectivity. On the positive tail, a negative total weight would eventually force work height below zero under repeated forward cycles, so the weight is positive.

For the negative left-infinite tail, a finite functional graph cannot support an infinite backward path outside directed cycles; sufficiently negative indices therefore lie on a cycle. A positive forward weight would force height below zero when followed backward through arbitrarily negative residuals, so its weight is negative.

If the two cycles shared a state, determinism of g would make them the same directed cycle with the same total weight, contradicting the opposite signs. □

Lemma 5.4 (Bounded-push cycle capacity)**.** *Assume $p \geq q$. Then*

$$L_+ \geq p, \qquad L_- \geq \max\left(q, \left\lceil \frac{p}{B-1} \right\rceil\right).$$

Proof. On the positive cycle, periodicity gives

$$h(r + L_+) = h(r) + c_+.$$

Hence, for r sufficiently positive,

$$h(r - pL_+) = h(r) - pc_+.$$

But 0^{L_+} realizes exactly that residual shift, remains on the deep positive tail, and one raw top rewrite can lower work height by at most one. Thus

$$pc_+ \leq L_+,$$

so $L_+ \geq p$.

On the negative cycle,

$$h(r + L_-) = h(r) - c_-.$$

Applying 0^{L_-} shifts the residual by $-pL_-$ and therefore raises height by pc_-. Each raw transition raises work height by at most $B - 1$, so

$$pc_- \leq L_-(B-1).$$

Applying 1^{L_-} shifts the residual by $+qL_-$ and lowers height by qc_-. One raw top rewrite can lower height by at most one, so

$$qc_- \leq L_-.$$

Since $c_- \geq 1$, the result follows. □

Theorem 5.5 (Exact raw bounded unary complexity)**.** *For every coprime $p, q \geq 1$ and every $B \geq 2$,*

$$\boxed{Q^{(B)}_{\text{raw,1work}} = M + \max\left(m, \left\lceil \frac{M}{B-1} \right\rceil\right).}$$

Proof. By symmetry assume $p \geq q$. The disjoint cycles of Theorem 5.3 and the capacity bounds of Theorem 5.4 require at least

$$p + \max\left(q, \left\lceil \frac{p}{B-1} \right\rceil\right)$$

states. Theorem 5.2 attains exactly that number. □

5.3 A separate three-symbol bounded-push optimum

The optimal three-symbol point can also be made replacement-at-most-two, but not within the one-transition raw category of Theorem 5.1. The correct interface is a normalized macro-step: one input-consuming move is followed, when necessary, by a uniquely forced terminating ε-burst, after which the stable residual representative has been restored.

Assume $p \geq q$ and use the $M = p$ state, three-symbol stable encoding from Appendix B. For $0 \leq u < q$, let

$$u^- = (u - p) \bmod q, \qquad d_u = \frac{u^- - u + p}{q},$$

and for a target remainder $v \in \mathbb{Z}_q$ put

$$\operatorname{pre}(v) = (v + p) \bmod q, \qquad L_v = d_{\operatorname{pre}(v)} - 1.$$

Since addition by p permutes $\mathbb{Z}_q$,

$$\sum_{v=0}^{q-1} L_v = \sum_{u=0}^{q-1} (d_u - 1) = p - q.$$

Introduce abstract suffix roles $H_{v,j}$, $1 \leq j \leq L_v$, meaning "push exactly j further copies of the negative work symbol B, then enter R_v." On a visible B their forced rules are

$$\delta(H_{v,1}, \varepsilon, B) = (R_v, BB), \qquad \delta(H_{v,j}, \varepsilon, B) = (H_{v,j-1}, BB) \quad (j > 1).$$

These are roles rather than new state names. Stable negative configurations use top B only in $R_0,\dots,R_{q-1}$, so the $p-q$ state/top cells

$$(R_i, B), \qquad q \le i < p,$$

are unused by stable semantics. Fix a bijection from the suffix roles onto those cells. A long negative input-0 write is replaced by one consuming $B \mapsto BB$ move into the appropriate suffix role; positive-to-negative boundary writes enter a suffix of the same chain. If $u_s = (s-p) \bmod q$ and $h_s = (u_s + p - s)/q$, then

$$d_{\mathrm{pre}(u_s)} = h_s + \left\lfloor \frac{s}{q} \right\rfloor,$$

so $h_s - 1 \le L_{u_s}$ and every required boundary suffix exists.

Theorem 5.6 (State-preserving push-at-most-two macro compiler)**.** *For every coprime p, q, there is a deterministic right-endmarker DPDA whose stable configurations realize the residual graph and satisfy*

$$\boxed{|Q| = M, \qquad |\Gamma| = 3, \qquad \max|\alpha| \le 2.}$$

Every nontrivial normalization burst is a uniquely forced finite ε-suffix chain. In particular, for every replacement bound $B \ge 2$, the minimum state count among deterministic recognizers with at most three total stack symbols is exactly M.

Proof. The role overlay above proves the case $p \ge q$. Every long negative write is split into one consuming push and a strictly descending suffix chain. The number of roles is exactly $p-q$, all fit into state/top cells unused by stable negative semantics, and those cells carry no competing input transition, so determinism is preserved. Boundary crossings use suffixes of the same chains, every chain terminates, and its macro-result equals the original stable residual transition. The case $q > p$ is symmetric. The baseline lower bound $|Q| \ge M$ from Theorem B.5 proves state optimality. For $B = 1$ stack height cannot increase, leaving only finitely many physical configurations, so no nonregular $L_{p,q}$ can be recognized. □

Corollary 5.7 (Critical raw unary push threshold)**.** *The unbounded one-work-symbol optimum $M + m = p + q$ is attained in the raw category if and only if*

$$\boxed{B \ge B_{\mathrm{crit}} := 1 + \left\lceil \frac{M}{m} \right\rceil.}$$

At $B = 2$, the exact raw unary state cost is $2M$.

Corollary 5.8 (Inverse raw unary state-to-push law)**.** *Fix a raw one-work-symbol state budget $S \ge M + m$. The least replacement bound is*

$$\boxed{B_{\min}(S) = \begin{cases} 2, & S \ge 2M, \\ 1 + \left\lceil \dfrac{M}{S-M} \right\rceil, & M + m \le S < 2M. \end{cases}}$$

Theorem 5.9 (Exact raw unary state–push staircase)**.** *Let*

$$Q_B = M + \max\!\left(m, \left\lceil \frac{M}{B-1} \right\rceil\right).$$

At the fixed total stack alphabet size $|\Gamma| = 2$, the Pareto-minimal raw pairs $(|Q|, B)$ are exactly

$$\boxed{\{(Q_B, B) : 2 \le B \le B_{\mathrm{crit}},\ B = 2 \text{ or } Q_B < Q_{B-1}\}.}$$

Proof. A plateau value is dominated by the same state count with the smaller replacement bound. Every point after B_{crit} is dominated by the critical-threshold point. At every strict drop, reducing B forces a larger state count by Theorem 5.5. Thus precisely the displayed breakpoints survive. □

Proposition 5.10 (The bounded three-symbol point is globally Pareto-minimal)**.** *In the broad deterministic permanent-bottom model, the resource triple*

$$\boxed{(M, 3, 2)}$$

is Pareto-minimal for $(|Q|, |\Gamma|, B)$*.*

Proof. It is attained by Theorem 5.6. A dominating triple cannot use fewer than M states by Theorem B.5; with M states it cannot use only two total stack symbols because Theorem 4.1 requires at least $p + q > M$ states there; and $B = 1$ cannot recognize the nonregular language. Hence no strict coordinate improvement can dominate $(M, 3, 2)$. □

The last proposition and Theorem 5.9 deliberately live in different categories. We do not combine them into an "exact raw three-resource frontier": the three-symbol bounded compiler uses forced ε-normalization, whereas Theorem 5.1 requires one physical input transition per residual edge.

6 Scope of the bounded result

The unrestricted theorem and the bounded raw theorem answer different questions. Theorem 4.1 quantifies over arbitrary recognizers but does not use a replacement bound. Theorem 5.5 measures the exact deformation caused by B after fixing a raw residual section.

For an arbitrary history-dependent one-work-symbol recognizer with bounded replacement, the present paper asserts only the proved envelope

$$p + q \le Q^{(B)}_{\mathrm{unrestricted,1work}} \le M + \max\left(m, \left\lceil \frac{M}{B-1} \right\rceil\right),$$

where the upper bound is supplied by a raw construction. No equality claim is made outside the typed raw category. This prevents a representation-relative cycle argument from being silently promoted to all deterministic pushdown realizations.

7 Examples and validation

For 2:1, the unrestricted state–stack frontier is

$$\{(3, 2), (2, 3)\}.$$

The bounded raw unary spectrum is summarized together with three further ratios in Table 1. The table displays the exact value from Theorem 5.5; repeated entries are plateaus and therefore do not all survive in the raw unary state–push staircase.

Table 1: Exact raw one-work-symbol state counts as the replacement bound varies.

ratio p:q	B_{crit}	$B = 2$	$B = 3$	$B = 4$	$B = 5$
2:1	3	4	3	3	3
3:2	3	6	5	5	5
5:2	4	10	8	7	7
7:2	5	14	11	10	9

The matching bounded raw construction was independently regression-checked for all coprime $1 \le q \le p \le 20$, all $2 \le B \le 10$, all residuals $|r| \le 200$, and both binary residual translations. The check verified exact target state/height, determinism of repeated state/top/input cells, and the replacement-length bound. No discrepancy was found. The computation is not used in any proof.

8 Related work and positioning

State hierarchies for deterministic pushdown automata and hierarchies based on limited pushdown alphabets are classical [1, 2, 3]. In particular, work of Meduna, Vrábel, Zemek, and Masopust studies stack-alphabet restrictions and fixed-state realtime deterministic pushdown models. One-counter automata are likewise classical, and recent descriptional-complexity work studies counter and pushdown memory usage [5]. The present problem fixes the natural commutative family $L_{p,q}$ and asks for exact parameter-dependent resource tradeoffs rather than constructing a new witness language for each hierarchy level.

The unrestricted state lower bound used as baseline is also distinct from the raw bounded theorem proved here. The former is language-level and representation-independent. The latter exploits the stronger residual-conjugacy interface to make physical push granularity exactly measurable. We do not claim that one-counter automata, state/stack hierarchies, or bounded stack updates are new notions.

9 Conclusion

For the rational binary balance family $L_{p,q}$, the state-only optimum does not exhaust the resource story. With the stack alphabet unrestricted, $M = \max(p, q)$ states suffice and are necessary. With only one work symbol, arbitrary deterministic ε-pushdown recognition requires exactly $p + q$ states. Therefore the complete unrestricted right-endmarker state/stack antichain is

$$\{(p + q, 2), (M, 3)\}.$$

When replacement length is charged inside the raw residual-conjugacy category, the unary branch unfolds into an exact finite staircase. The formula

$$M + \max\left(m, \left\lceil \frac{M}{B-1} \right\rceil\right)$$

gives the exact state cost and yields the critical push threshold $1+\lceil M/m \rceil$; after deleting plateaus it gives the exact raw unary $(|Q|, B)$ staircase at fixed $|\Gamma| = 2$. Separately, the normalized macro-step compiler attains the globally Pareto-minimal bounded point $(M, 3, 2)$. These two bounded statements deliberately remain in different categories. Together with the unrestricted two-resource antichain, they separate what is forced by the language from what is forced by a declared physical representation contract.

A Explicit bounded unary transition schema

For completeness, we record the full transition schema underlying Theorem 5.2. Assume $p \ge q$, put

$$b = \max\left(q, \left\lceil \frac{p}{B-1} \right\rceil\right),$$

and use states

$$P_0, \ldots, P_{p-1}, \qquad N_0, \ldots, N_{b-1}.$$

The stable encoding is

$$\eta(r) = \begin{cases} (P_s, A^k Z_0), & r = pk + s \geq 0,\ 0 \leq s < p, \\ (N_u, A^{h-1} Z_0), & r = u - bh < 0,\ 0 \leq u < b,\ h \geq 1. \end{cases}$$

For $0 \leq s < p$, define

$$s^+ = (s+q) \bmod p, \qquad c_s = \frac{s+q-s^+}{p} \in \{0,1\},$$

and

$$u_s = (s-p) \bmod b, \qquad h_s = \frac{u_s + p - s}{b}.$$

Since $b \geq \lceil p/(B-1) \rceil$, one has $1 \leq h_s \leq B-1$. The positive and positive-boundary rules are

$$\begin{aligned} \delta(P_s, A, 0) &= (P_s, \varepsilon), \\ \delta(P_s, A, 1) &= (P_{s^+}, A^{c_s+1}), \\ \delta(P_s, Z_0, 1) &= (P_{s^+}, A^{c_s} Z_0), \\ \delta(P_s, Z_0, 0) &= (N_{u_s}, A^{h_s-1} Z_0). \end{aligned}$$

For $0 \leq u < b$, define

$$u^- = (u-p) \bmod b, \qquad d_u = \frac{u^- - u + p}{b},$$

and

$$u^+ = (u+q) \bmod b, \qquad e_u = \frac{u+q-u^+}{b} \in \{0,1\}.$$

Because $0 \leq u < b \leq p$, one has $1 \leq d_u \leq \lceil p/b \rceil \leq B-1$. On a non-bottom negative stack,

$$\delta(N_u, A, 0) = (N_{u^-}, A^{d_u+1}),$$

and

$$\delta(N_u, A, 1) = \begin{cases} (N_{u^+}, A), & e_u = 0, \\ (N_{u^+}, \varepsilon), & e_u = 1. \end{cases}$$

At the bottom,

$$\delta(N_u, Z_0, 0) = (N_{u^-}, A^{d_u} Z_0),$$

while

$$\delta(N_u, Z_0, 1) = \begin{cases} (N_{u^+}, Z_0), & e_u = 0, \\ (P_{u^+}, Z_0), & e_u = 1. \end{cases}$$

The second branch is well typed because $u^+ < q \leq p$. Finally, the right endmarker is accepted only from (P_0, Z_0).

Direct substitution gives

$$\delta(\eta(r), 0) = \eta(r-p), \qquad \delta(\eta(r), 1) = \eta(r+q)$$

for every $r \in \mathbb{Z}$. The longest work-top replacement has length $d_u + 1 \leq B$, and the longest bottom replacement has total length $d_u + 1 \leq B$. The construction is therefore raw, deterministic, ε-free, and B-bounded.

When $B \geq 1 + \lceil p/q \rceil$, the modulus collapses to $b = q$, and this schema is exactly the $p+q$-state unbounded one-work-symbol construction used to attain Theorem 4.1.

B Self-contained baseline state theorem and interface

This appendix supplies the representation-independent theorem used in (1) and proves the pump/exposure interface invoked in the main text. We keep the paper's convention that the visible stack top is written on the *left*.

B.1 Residual quotient rigidity and unary escape

For $w \in \{0,1\}^*$, put $\rho(w) = qN_1(w) - pN_0(w)$. Since $\gcd(p,q) = 1$, every integer occurs as the residual of some binary word: if $qx_0 - py_0 = t$, then $x = x_0 + pk$, $y = y_0 + qk$ are nonnegative for all sufficiently large k.

Lemma B.1 (Residual right-quotient rigidity)**.** *For binary prefixes u, v,*

$$u^{-1}L_{p,q} = v^{-1}L_{p,q} \iff \rho(u) = \rho(v).$$

Consequently two consumed prefixes reaching the same physical configuration of a deterministic recognizer have the same residual.

Proof. Equal residuals clearly give equal accepting continuation sets. Conversely, choose a continuation of residual $-\rho(u)$; it accepts after u and, if the residuals differ, rejects after v. A deterministic physical configuration has one continuation language, so it cannot carry two residual labels. ☐

Consider the live computation over 1^ω. Every finite unary prefix has a finite accepting extension, so the computation must consume arbitrarily many input symbols and cannot become trapped in a deterministic ε-cycle. No full raw configuration can recur: recurrence after different consumed-input positions violates Theorem B.1, while recurrence at the same position creates an infinite ε-cycle. Hence stack height eventually leaves every finite bound. The integer-valued height sequence therefore has suffix-minimum positions at arbitrarily large heights.

Lemma B.2 (Protected pump, left-top notation)**.** *There are a state r, a visible symbol X, an integer $c \geq 1$, a nonempty word U, and a lower context η such that*

$$(r, X\eta) \overset{1^c}{\Longrightarrow} (r, XU\eta),$$

the macro never drops below the initial height, and the raw transition sequence is unchanged if the hidden lower context η is replaced by another context. Hence it iterates:

$$(r, XU^k\eta) \overset{1^c}{\Longrightarrow} (r, XU^{k+1}\eta), \qquad k \geq 0.$$

Proof. Among suffix minima of arbitrarily large height, some state/top mode (r, X) occurs twice, the later occurrence higher than the earlier. Between them the computation never drops below the earlier suffix minimum. Thus the lower context cannot be exposed; at the second occurrence the extra material lies immediately below the same visible top X, giving $XU\eta$ with $U \neq \varepsilon$. Every transition sees only the active portion above the hidden context, proving context independence. If no input symbol were consumed, the same height-increasing deterministic ε-macro would repeat forever, contradicting liveness. Therefore $c \geq 1$. ☐

Put $H_k = U^k\eta$. Starting from (r, XH_k), consume $j \in \{0, \dots, p-1\}$ further ones and all forced ε-moves until the next-input boundary. Because the future computation remains above H_k, for fixed j the resulting state and upper stack word are independent of k; write the configuration as

$$D_{k,j} = (q_j, W_jH_k), \qquad \Theta(D_{k,j}) = A + kcq + jq.$$

Follow 0^ω from $D_{k,j}$. Until H_k is exposed, the relative raw history above the cut is independent of k. Thus either every H_k is first exposed after one fixed number z_j of consumed zeros in one fixed state s_j, or no H_k is ever exposed.

Lemma B.3 (Fixed-continuation stack regularity)**.** *Fix a state s and a finite continuation v, including the right endmarker when present. The set of stacks from which v leads to acceptance is regular.*

Proof. Adjoin the input position in v to finite control. The problem becomes predecessor reachability of a regular target set in an ordinary pushdown system. Regular sets of pushdown configurations are closed under predecessor reachability by the standard saturation theorem [4]. Restriction to the fixed state gives a regular language of stacks. □

Lemma B.4 (Permanent deferral is impossible)**.** *Every positive phase exposes its protected cut; the same holds on the symmetric negative ray.*

Proof. Assume one positive phase never exposes H_k. The relative 0^ω computation is live, nonrecurrent, and therefore height-escaping. Applying Theorem B.2 with 0 in place of 1 yields a second context-independent pump above the hidden cut, consuming $d \geq 1$ zeros per iteration. For one fixed state this gives a two-parameter reachable family of stacks, obtained by independently repeating the first and second pump blocks, whose residuals are

$$B + kcq - \ell dp.$$

Choose one fixed continuation with residual $-B$. Correct recognition then accepts exactly when

$$kcq = \ell dp.$$

The inverse image of the regular stack language from Theorem B.3 under the two-block morphism $a^k b^\ell \mapsto$ "first-pump block^k second-pump block$^\ell$" would be regular. Its intersection with $a^* b^*$ is instead the nonregular positive linear diagonal above, a contradiction. □

At first exposure, if $s_j = s_t$ for $0 \leq j < t < p$, the common stack H_k would carry residual difference

$$q(j - t) - p(z_j - z_t) = 0.$$

Coprimality would force $p \mid (j - t)$, impossible. Hence the p positive exposure states are distinct. Reversing the roles of $0, 1$ and p, q gives q distinct negative phases.

Theorem B.5 (Representation-independent state bound)**.** *Every deterministic ε-pushdown recognizer of $L_{p,q}$ in the regular-target model satisfies*

$$|Q| \geq \max(p, q).$$

Proof. The positive construction gives p distinct states and the symmetric negative construction gives q distinct states. Either count alone is a lower bound, so $|Q| \geq \max(p, q)$. □

The argument used no canonical residual encoding, homogeneous stack tail, bounded replacement length, or realtime assumption. The main-text interface lemmas Theorems 2.2 and 2.3 are precisely the protected-pump and exposure conclusions just proved.

B.2 Matching three-symbol right-endmarker construction

For completeness we also give the attainment used by Theorem 4.2. Assume $p \geq q$; the other orientation is obtained by exchanging 0 and 1, p and q, and the two work symbols. Use states

$$R_0, \ldots, R_{p-1}$$

and stack alphabet $\{A, B, Z_0\}$. For $r \geq 0$, write $r = pk + s$, $0 \leq s < p$, and set

$$\zeta(r) = (R_s, A^k Z_0).$$

For $r < 0$, write uniquely $r = u - qh$, $0 \le u < q$, $h \ge 1$, and set

$$\zeta(r) = (R_u, B^h Z_0).$$

Put

$$s^+ = (s+q) \bmod p, \qquad c_s = \frac{s+q-s^+}{p} \in \{0,1\},$$

$$u_s = (s-p) \bmod q, \qquad h_s = \frac{u_s+p-s}{q} \ge 1,$$

and, for $0 \le u < q$,

$$u^- = (u-p) \bmod q, \qquad d_u = \frac{u^- - u + p}{q} \ge 1.$$

With top written on the left, define

$$\delta(R_s, A, 0) = (R_s, \varepsilon), \qquad \delta(R_s, A, 1) = (R_{s^+}, A^{c_s+1}),$$

$$\delta(R_s, Z_0, 1) = (R_{s^+}, A^{c_s} Z_0), \qquad \delta(R_s, Z_0, 0) = (R_{u_s}, B^{h_s} Z_0),$$

and

$$\delta(R_u, B, 1) = (R_u, \varepsilon), \qquad \delta(R_u, B, 0) = (R_{u^-}, B^{d_u+1}).$$

No ε-move is used. The right endmarker is accepted only from (R_0, Z_0).

Proposition B.6 (Three-symbol attainment)**.** *The displayed machine satisfies, for every $r \in \mathbb{Z}$,*

$$\delta(\zeta(r), 0) = \zeta(r-p), \qquad \delta(\zeta(r), 1) = \zeta(r+q).$$

Hence it recognizes $L_{p,q}$ with $p = \max(p,q)$ states and three total stack symbols.

Proof. For $r = pk+s \ge 0$, input 0 pops one A when $k \ge 1$; at $k = 0$, the identity $s-p = u_s - qh_s$ gives the negative boundary rule. The carry identity $s+q = pc_s + s^+$ proves the input-1 rule both on work top and at the bottom. For $r = u - qh < 0$, input 1 pops one B, while $u - p = u^- - qd_u$ gives the input-0 update. Thus every binary step performs the correct residual translation, and endmarker acceptance occurs exactly at residual zero. $\square$

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